

Laser Heating Beamtime Pre-Experiment Questionnaire

General

1. Beamtime Dates

2. Radiological Beamtime?

No

Yes

Partially

3. What technique(s) are we trying to use?

Standard DAC (no LH)

Laser-heated DAC

Membrane-driven compression

Dynamic DAC (dDAC / time-resolved)

Resistively heated

Single-crystal / Multi-grain DAC

Toroidal DAC

Other (please specify):

4. What DAC types will you be bringing?

Standard symmetric DAC (Princeton type)

BX-90

Membrane-driven DAC

Dynamic DAC (dDAC)

DAC in vacuum jacket

Custom / in-house DAC

Need a custom holder for DAC?

Other (please specify):

Beam Specs

5. Sample chamber sizes (μm) (please list)

6. Required X-ray focus, FWHM ($\mu\text{m} \times \mu\text{m}$)

Typical focus ($\mu\text{m} \times \mu\text{m}$): 1.5×1.5

7. Preferred X-ray energy (keV)

▼

Default energy (keV): 29.2. 16-ID-B uses 29.2 keV for 99% of its experiments, note that requesting a different energy will add some setup time for your beamline scientist.

8. Do you anticipate needing to defocus or change beam size during experiment?

No Yes

If yes, please describe:

Mirrors

9. Which mirror will be used? (default: LKB)

SKB LKB

10. Isolation mode? (Only for radiological beamtime)

No Yes

Expected energy, mirror, or isolation mode changes during beamtime:

Beamtime Daily Plan

Day 1:

Day 2:

Day 3:

Day 4:

Day 5+:

Capabilities

11. Online ruby system?

No Yes

12. Other high temperature method(s)

Resistive heating

Please describe:

13. Do you need any of the following?

Vacuum pump Chiller

Thermocouple interface

Other (please specify):

14. How do you want to heat?

Square modulated Continuous

Please describe: